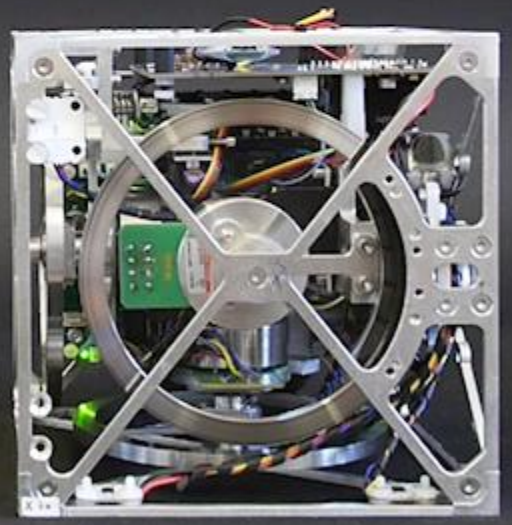
On the hardness of learning in dynamical systems

Michael Muehlebach

Vienna University of Economics and Business
Max Planck Institute for Intelligent Systems
https://lds.is.tuebingen.mpg.de/



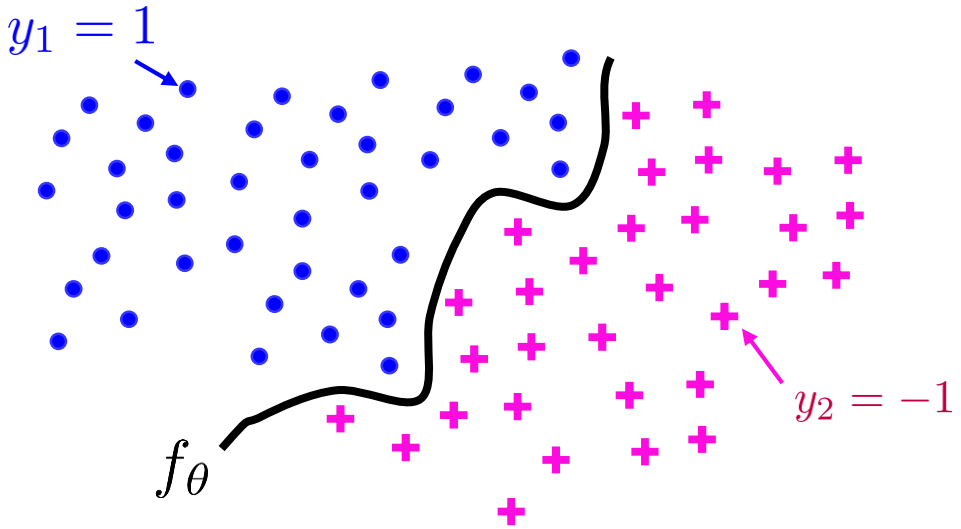


M. Muehlebach, R. D' Andrea, "*Nonlinear analysis and control of a reaction wheel-based 3D inverted pendulum*", IEEE Transactions on Control Systems Technology, 2017

Reinforcement vs. supervised learning

Supervised learning

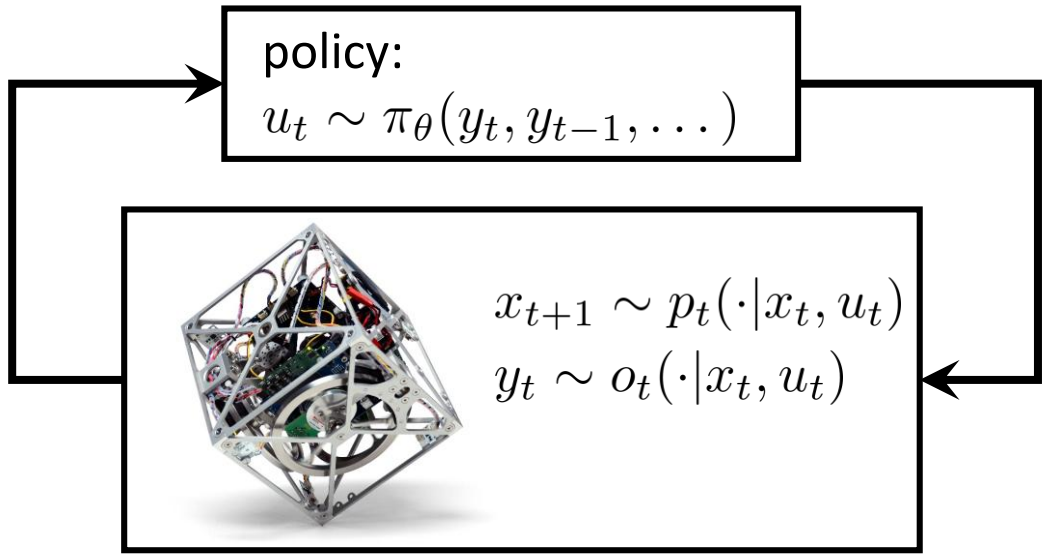
$$(x_i, y_i)_{i=1}^{T} \sim p(X, Y)$$



→ No feedback!

→ We never change data-generating process!

Reinforcement learning



→ Policy changes distribution over state!

→ Care about counterfactuals/long-term consequences!

[Jordan'18, Schölkopf'19, Bottou'23, Karpathy'23, LeCun'24, Sutton'25,…]

Reinforcement vs. supervised learning

Supervised learning

Reinforcement learning

When is reinforcement no harder than supervised learning?

Under structural assumptions, e.g., ***benign distribution shift, identifiability/persistence of excitation, contractive dynamics***, we can match supervised-learning rates even though we are ***intervening in a dynamical system***!

→ We never change data generating process!

→ Care about counterfactuals/long-term consequences!

[Jordan'18, Schölkopf'19, Bottou'23, Karpathy'23, LeCun'24, Sutton'25,…]

Meta-theorem:

Supervised learning has excess risk $\mathcal{O}(\sqrt{\log(m)/T})$, regret $\mathcal{O}(\sqrt{\log(m)T})$.*

Reinforcement learning **has the same rate** if:

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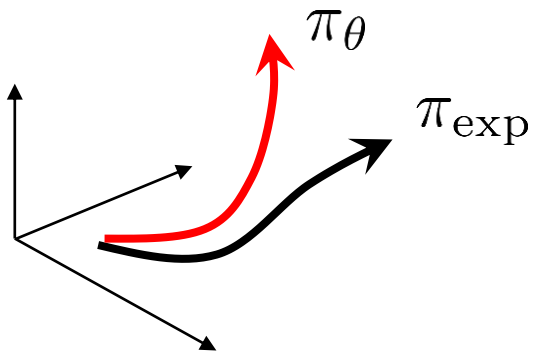
*well known, see, e.g., [Robbins'51, Wolfowitz'54, Vapnik'95, Cesa-Bianchi'06, Bach'24, …]

Imitation learning/offline RL

- Given: static dataset $(y_1, u_1), \ldots, (y_T, u_T)$ of trajectories obtained from applying $\pi_{\text{exp}} : \mathcal{Y} \to \mathcal{U}$
- Goal: learn policy $\pi_\theta : \mathcal{Y} \to \mathcal{U}$ that imitates (or surpasses) expert

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- What can go wrong?

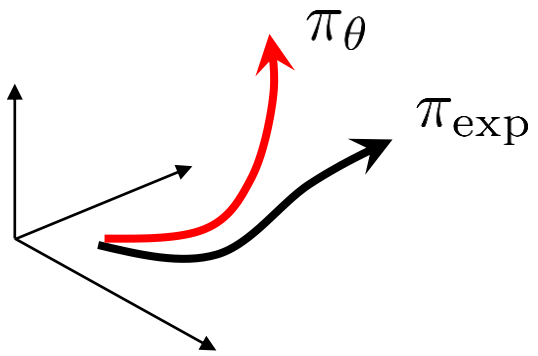


compounding error

[Ross'11, Zhang'25, Simchowitz'25, …]

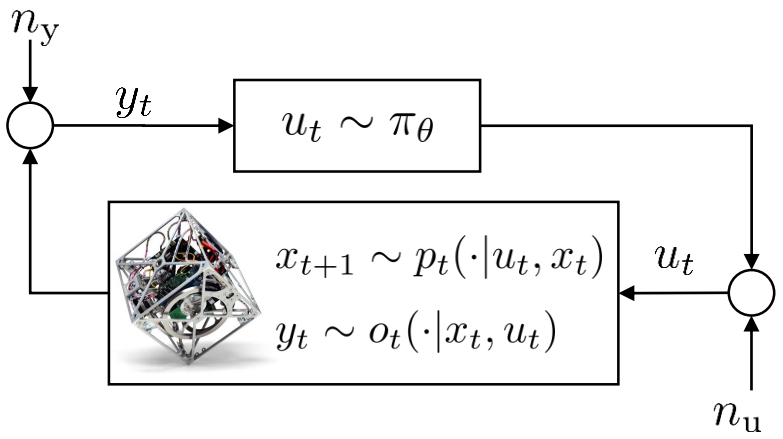
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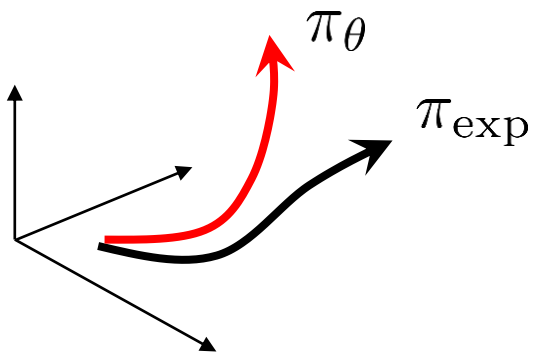


distribution shift

[Bode'45,Wiener'48,Anderson'86,…]

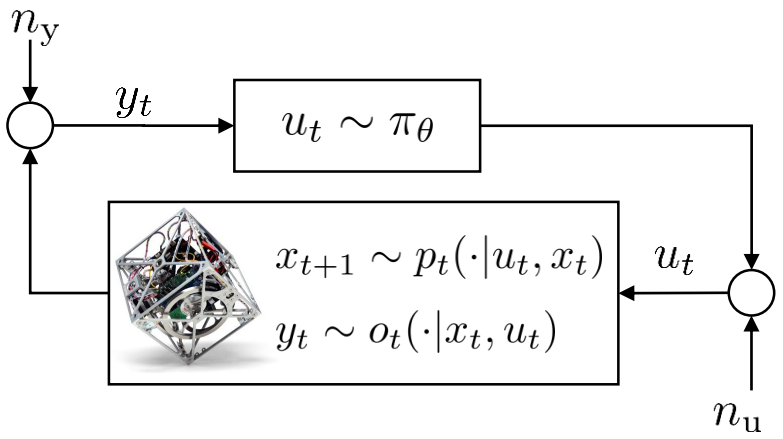
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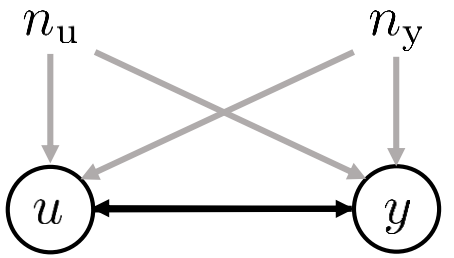
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[Ross'11, Zhang'25, Simchowitz'25, …]



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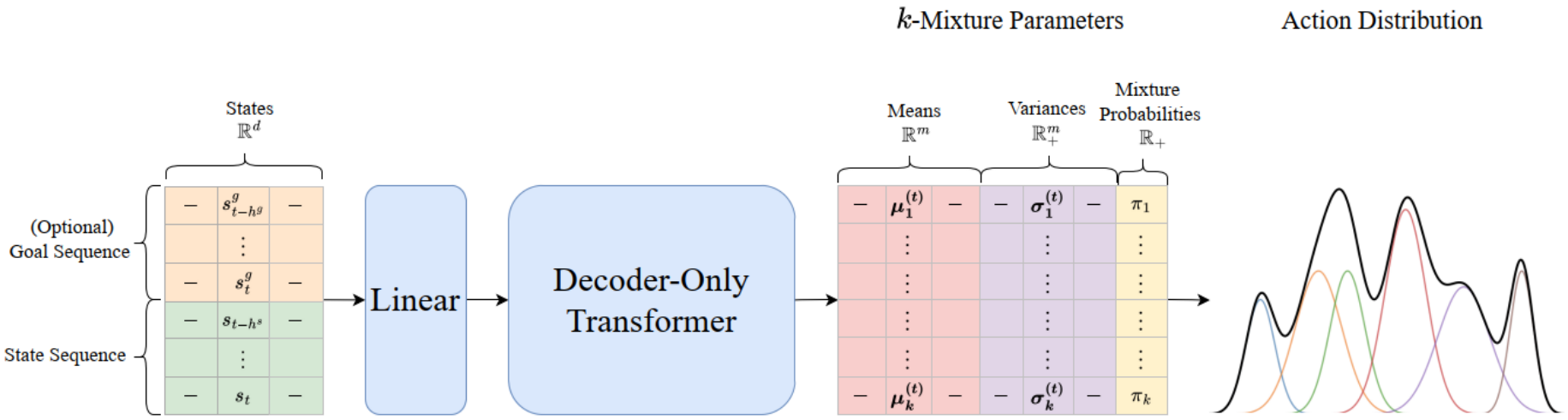
simultaneity bias

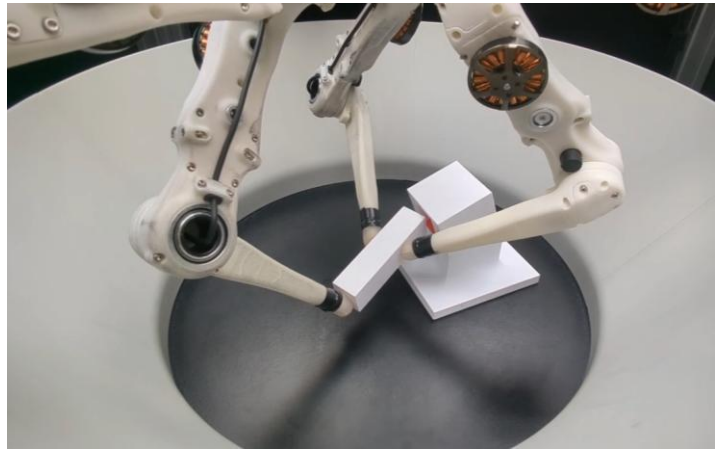
[Merton'68, Ljung'87, …]

Imitation learning/offline RL

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- Engineering challenges: expressiveness of π_θ, low latency, likelihood computation, varying embodiments, incorporating constraints

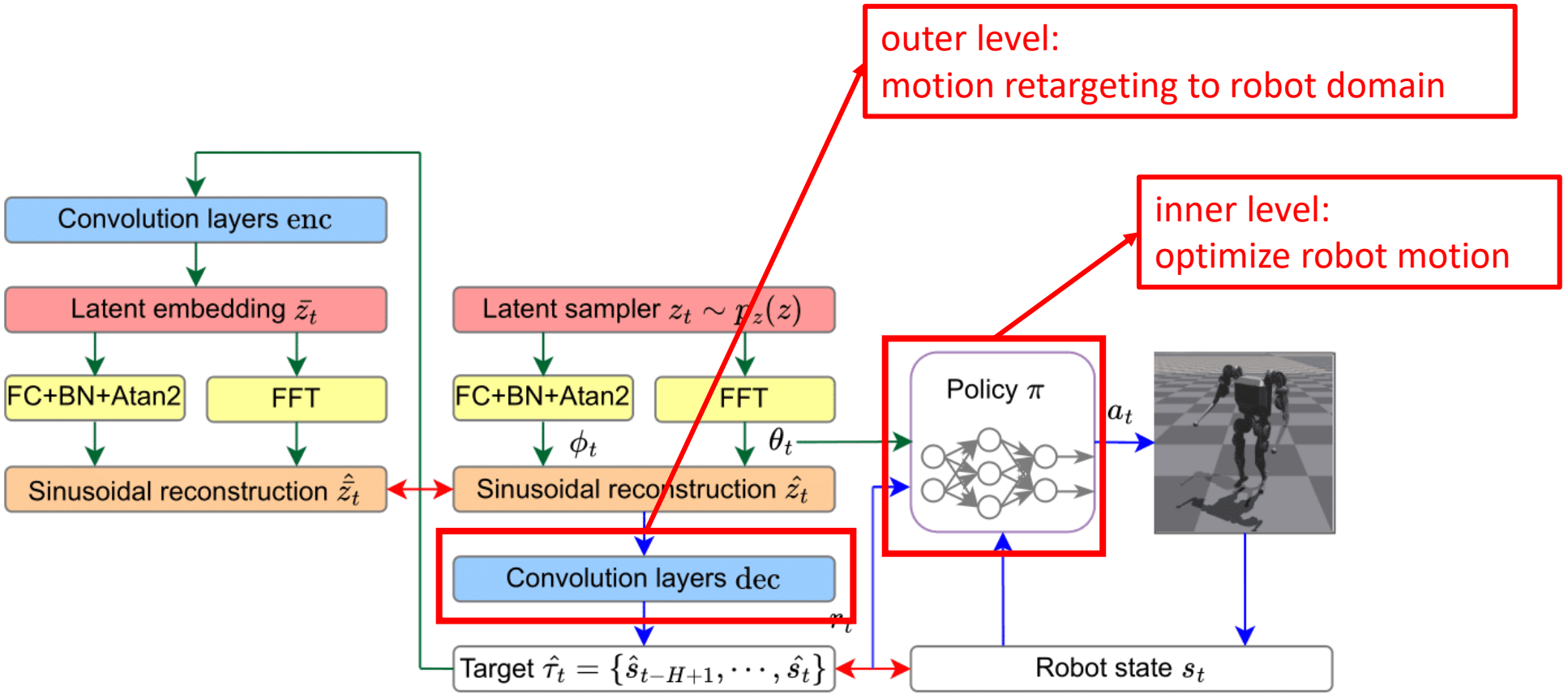
Quantization-Free Action Transformers





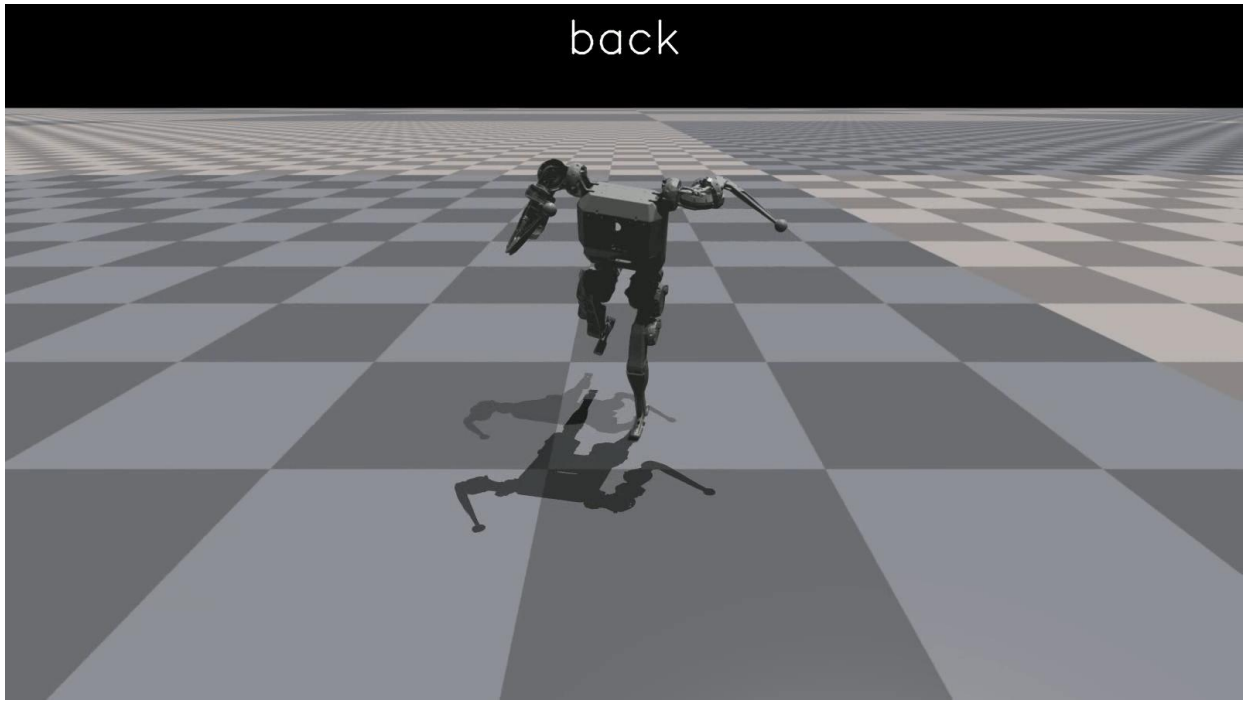
Z. Sheebaelhamd, M. Tschannen, M. Muehlebach, C. Vernade, "*Quantization-Free Autoregressive Action Transformers*", Advances in Neural Information Processing Systems, 2025

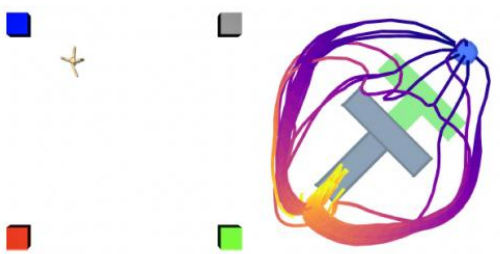
Bilevel optimization for embodiment change

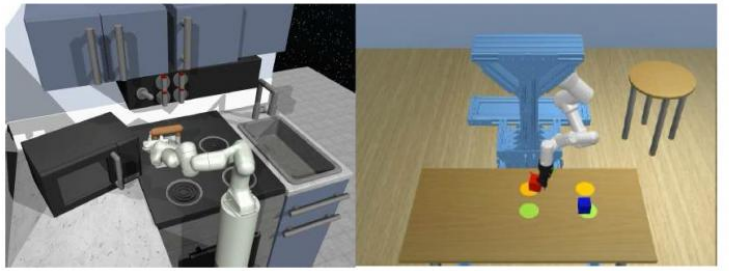


W. Zhao, Y. Zhao, J. Pajarinen, M. Muehlebach, "*Bi-Level Motion Imitation for Humanoid Robots*", CoRL 2024

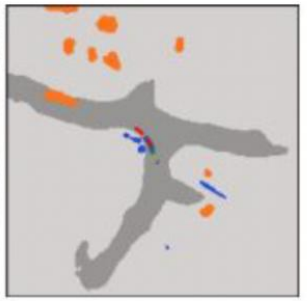
Imitation learning











— Unconditional Tasks —						
Environment	Metric	BeT	DiffPolicy-C	DiffPolicy-T	VQ-BeT	Q-FAT
PushT	Final IoU (·/1)	0.39	0.73	0.74	0.78	**0.80**
Image PushT		0.01	0.66	0.45	0.68	**0.69**
Kitchen	Goals (·/4)	3.07	2.62	3.44	3.66	**3.84**
Image Kitchen		2.48	3.11	3.01	2.98	**3.15**
Multimodal Ant		2.73	3.12	2.90	3.22	**3.30**
UR3 BlockPush	Goals (·/2)	1.59	1.83	1.82	1.84	**1.99**
BlockPush		1.67	0.47	**1.93**	1.79	1.77

— Conditional Tasks —						
Environment	Metric	C-BeT	C-BESO	CFG-BESO	VQ-BeT	Q-FAT
Cond. Kitchen	Goals (·/4)	0.15	3.75	3.47	**3.78**	**3.78**
Cond. UR3 BlockPush	Goals (·/2)	0.00	1.14	0.92	1.94	**1.96**

W. Zhao, Y. Zhao, J. Pajarinen, M. Muehlebach, "*Bi-Level Motion Imitation for Humanoid Robots*", Conf. on Robot Learning, 2024

Z. Sheebaelhamd, M. Tschannen, M. Muehlebach, C. Vernade, "*Quantization-Free Autoregressive Action Transformers*", Advances in Neural Information Processing Systems, 2025

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Online reinforcement learning

- Parametric class:

$$F = \{f^\theta \mid \theta \in \Omega \subset \mathbb{R}^p\}$$

$$x_{k+1} = f^\theta(x_k, u_k) + n_k, \quad n_k \sim \mathcal{N}(0, \sigma^2 I)$$

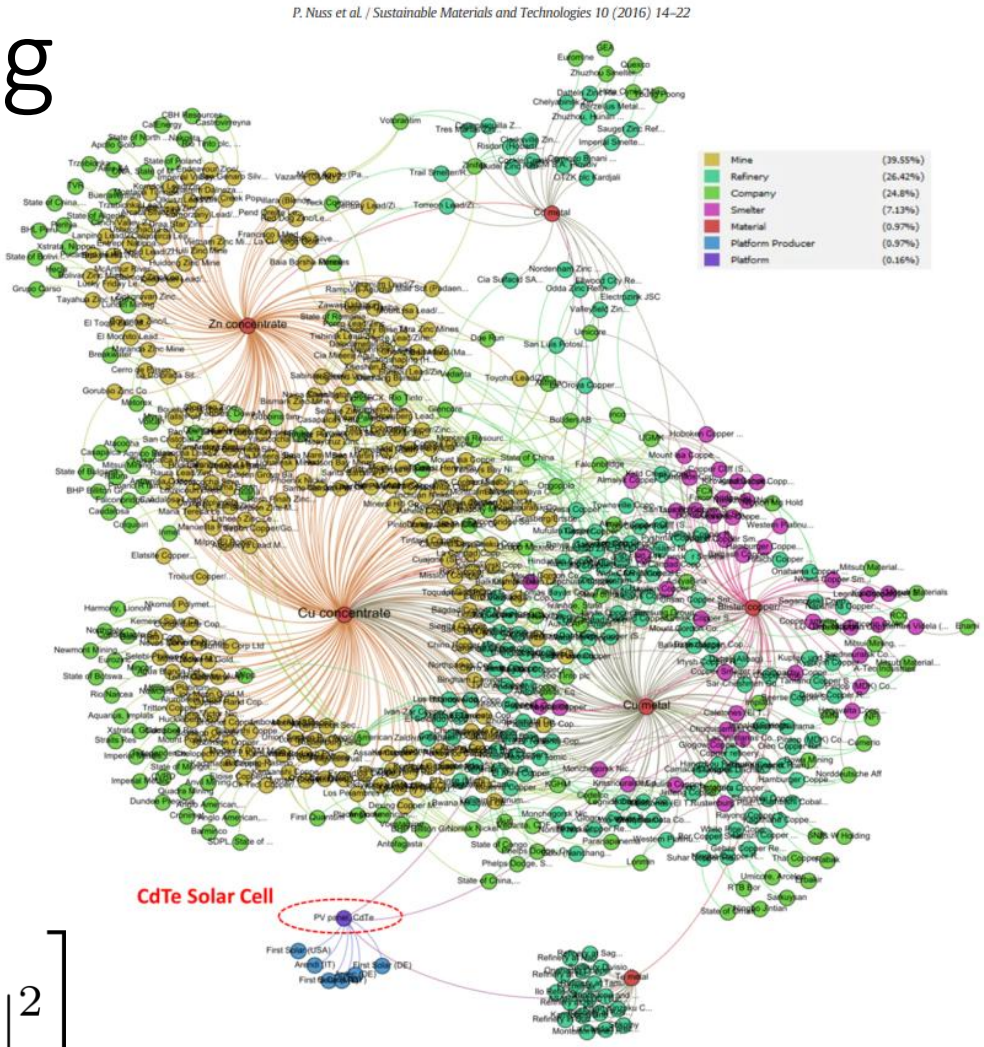
- Finite class of models:

$$F = \{f^1, \ldots, f^m\}$$

$$x_{k+1} = f^i(x_k, u_k) + n_k, \quad n_k \sim \mathcal{N}(0, \sigma^2 I)$$

- Regret:

$$\mathrm{E}\left[\sum_{k=0}^{T-1} l(x_k, u_k)\right] - \inf_{f \in F} \mathrm{E}\left[\sum_{k=0}^{T-1} \gamma(f)|n_k|^2\right]$$



[Nuss et. al., 2016]

M. Muehlebach, Z. He, M. I. Jordan, "*The Sample Complexity of Online Reinforcement Learning: A Multi-model Perspective*", International Conference on Learning Representations, 2026

Prior works

- Online Linear Quadratic Regulator $\tilde{\Theta}(\sqrt{d_{\mathrm{u}}^2 d_{\mathrm{x}} T})$
  - → Least-squares and optimal control (regret): [Dean et al., 2018, Simchowitz et al., 2020]
  - → Policy gradient methods (convergence): [Fazel et al. 2019, Mohammadi et al. 2021]
- Posterior Sampling Reinforcement Learning $\tilde{\mathcal{O}}(\sqrt{\tau|A||S|T})$ - episodic/tabular
  - → Episodic finite state-action systems: [Osband et al., 2013, Abbasi-Yadkori et al., 2015]
  - → Mostly Bayesian regret, episodic, might need mixing assumptions
- Optimism in the Face of Uncertainty $\tilde{\mathcal{O}}(\sqrt{\tau|S||A|T})$ - episodic/tabular
  - → Tabular setting: [Auer et al., 2006, Jaksch et al., 2010, Azar et al., 2017]
  - → Mostly episodic and tabular, exploration bonus might be difficult to compute
- Online Optimal Control $\tilde{\mathcal{O}}(\sqrt{T})$ - regret vs. linear / disturbance-feedback
  - → Disturbance feedback policy: [Hazan et al., 2022, Tsiamis et al., 2023]
  - → Nonlinear dynamics [Kakade et al., 2020, Boffi et al., 2021, Lin et al., 2024]

→ Key open question: non-episodic regret for nonlinear systems!

Algorithm

Algorithm 1 Online reinforcement learning (S1)

Require: $F := \{f^1, \dots, f^m\}$, η, M, $\{\sigma_{\mathrm{u}k}^2\}_{k=1}^{\infty}$

compute $\{\mu^1, \dots, \mu^m\}$ // e.g. by dynamic programming

for $k = 1, \dots$ **do**

// every Mth step

if $\mathrm{mod}(k-1, M) = 0$ **then**

$s_k^i \leftarrow \sum_{j=1}^{k-1} |x_{j+1} - f^i(x_j, u_j)|^2$ //one-step pred. error

$i_k \overset{\text{i.i.d.}}{\sim} \exp(-\eta s_k^i)$ //sample from $e^{-\eta s_k^i}$/norm

else

$i_k = i_{k-1}$ //stay with i_{k-1}

end if

//follow policy i_k and add excitation

$u_k = \mu^{i_k}(x_k) + n_{\mathrm{u}k}, \quad n_{\mathrm{u}k} \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma_{\mathrm{u}k}^2 I)$

end for

Key elements:

certainty-equivalence:

offline computation → e.g. with proximal policy optimization

online computation → e.g. with nonlinear model predictive control

M. Muehlebach, Z. He, M. I. Jordan, "*The Sample Complexity of Online Reinforcement Learning: A Multi-model Perspective*", International Conference on Learning Representations, 2026

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M. Muehlebach, Z. He, M. I. Jordan, "*The Sample Complexity of Online Reinforcement Learning: A Multi-model Perspective*", International Conference on Learning Representations, 2026

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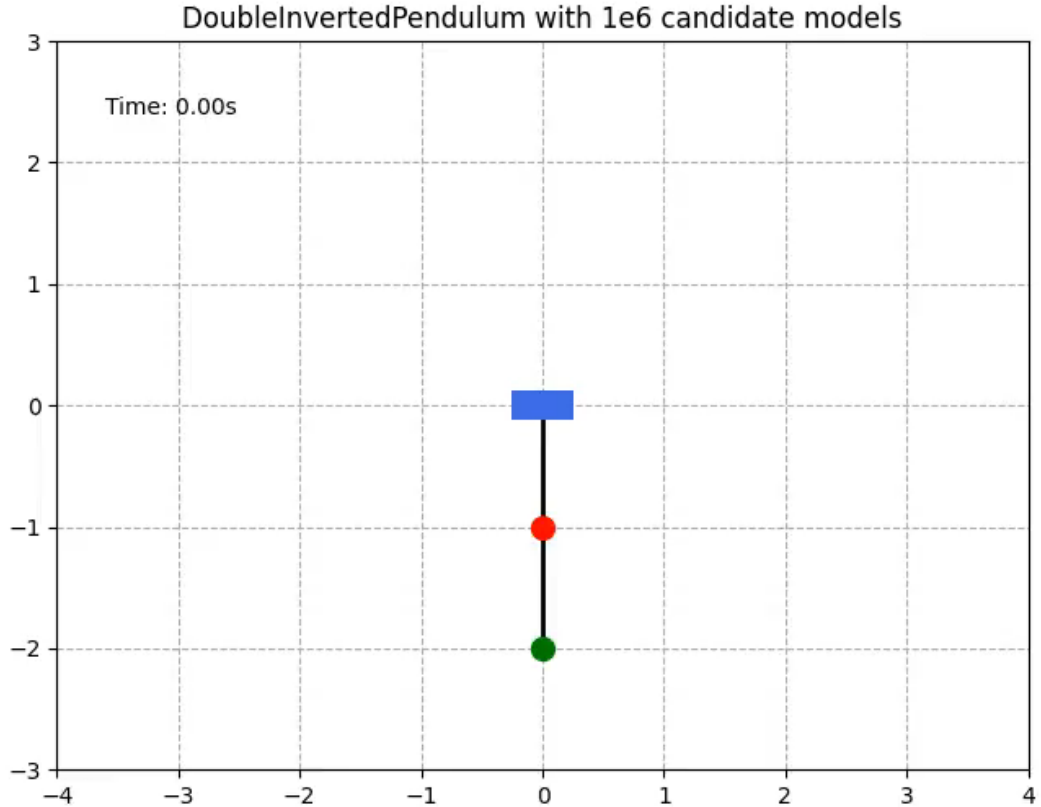
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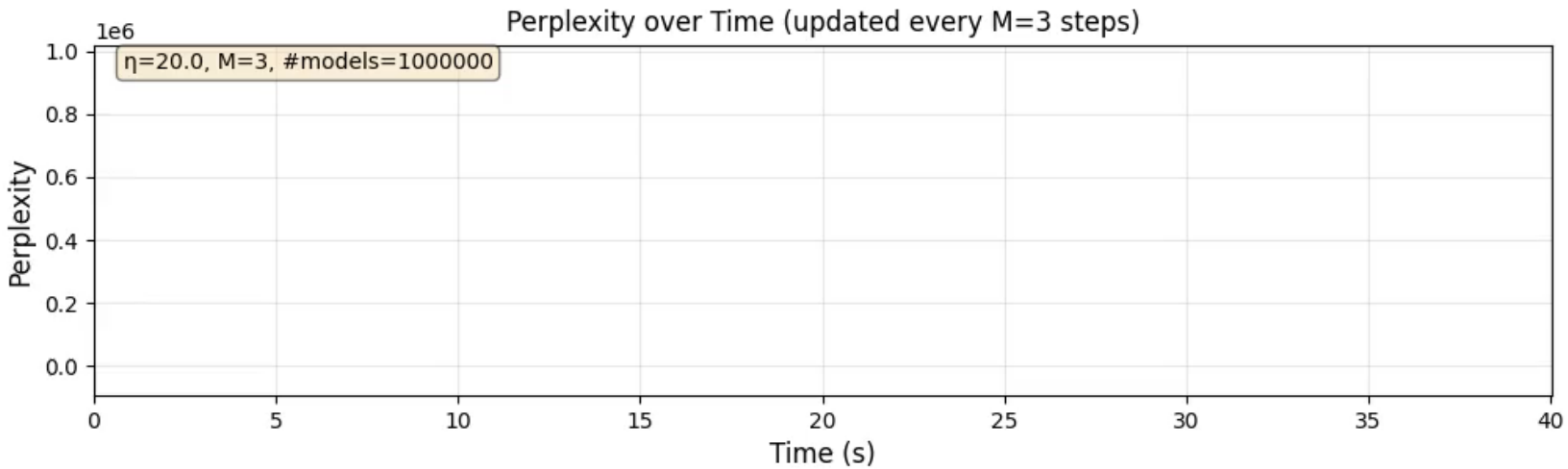
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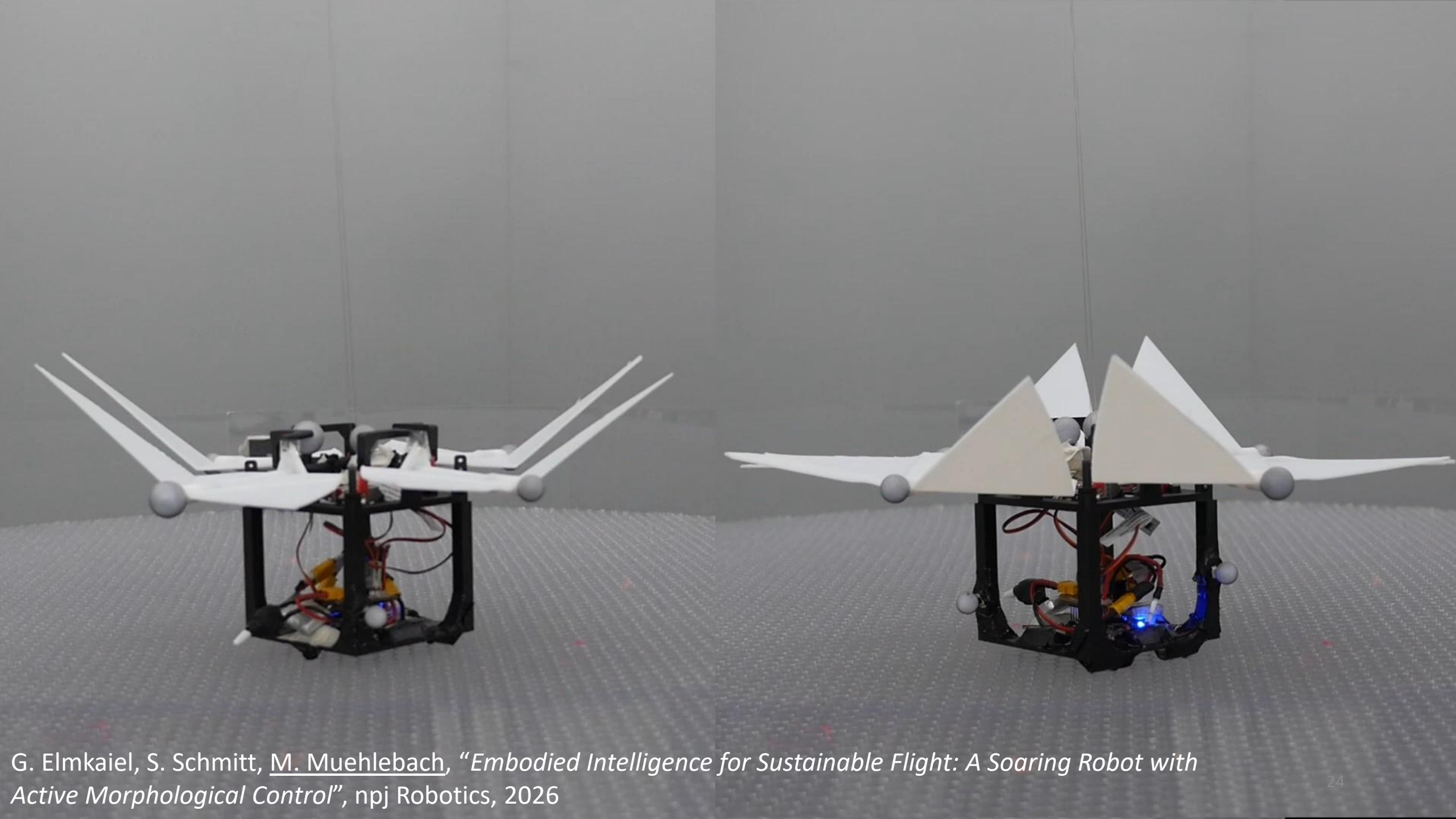
exploration/excitation

M. Muehlebach, Z. He, M. I. Jordan, "*The Sample Complexity of Online Reinforcement Learning: A Multi-model Perspective*", International Conference on Learning Representations, 2026


DoubleInvertedPendulum with 1e6 candidate models
Time: 0.00s
3
2
1
0
−1
−2
−3
−4
−3
−2
−1
0
1
2
3
4


Perplexity over Time (updated every M=3 steps)
1e6
η=20.0, M=3, #models=1000000
Perplexity
1.0
0.8
0.6
0.4
0.2
0.0
0
5
10
15
20
25
30
35
40
Time (s)

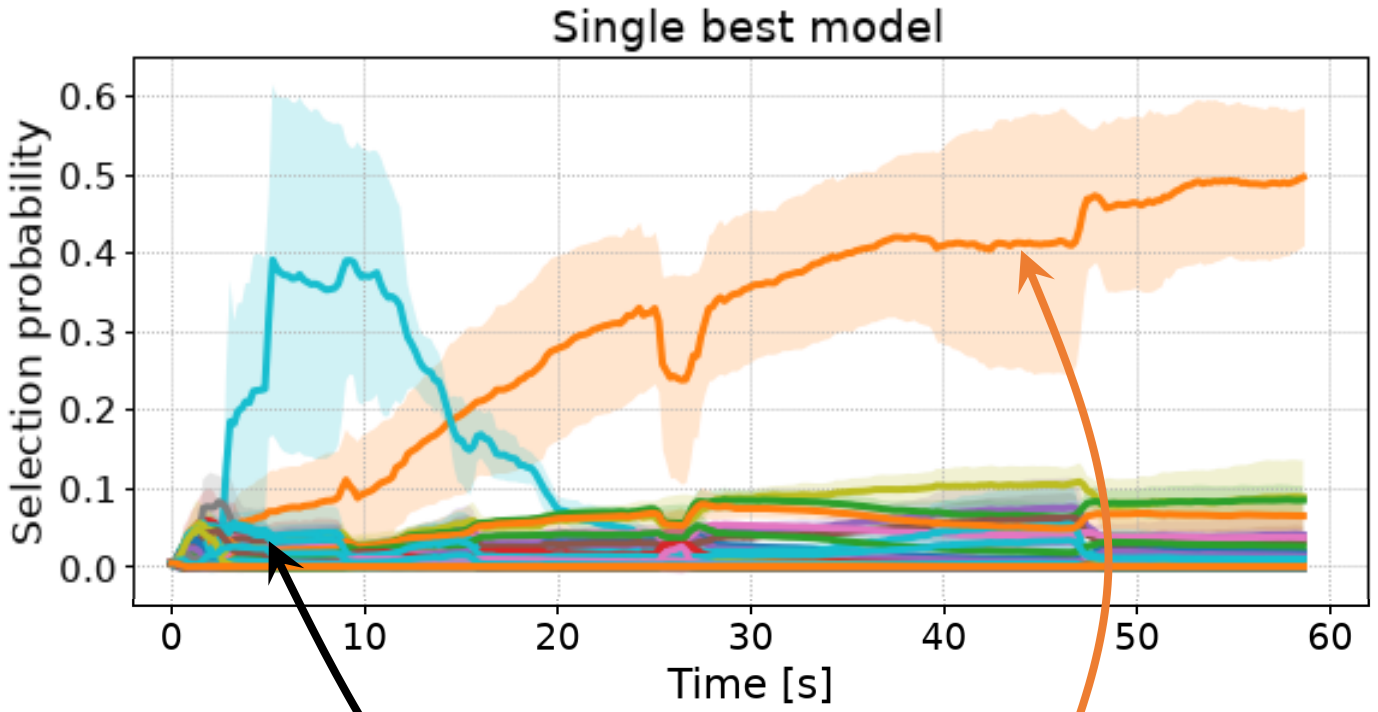




G. Elmkaiel, S. Schmitt, M. Muehlebach, “*Embodied Intelligence for Sustainable Flight: A Soaring Robot with Active Morphological Control*”, npj Robotics, 2026



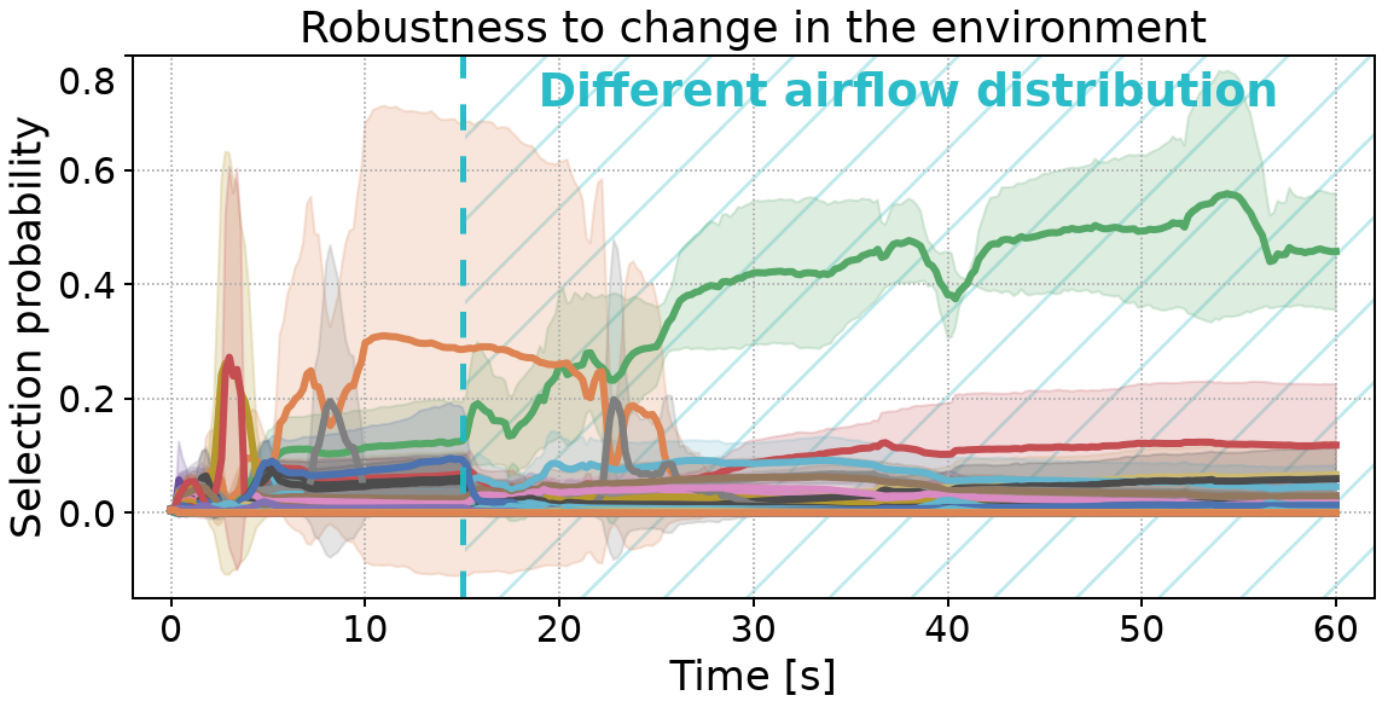
G. Elmkaiel, M. Zaric, M. Muehlebach, "*Fast online reinforcement learning on real robots*", Conference on Robot Learning, 2026 (in review)





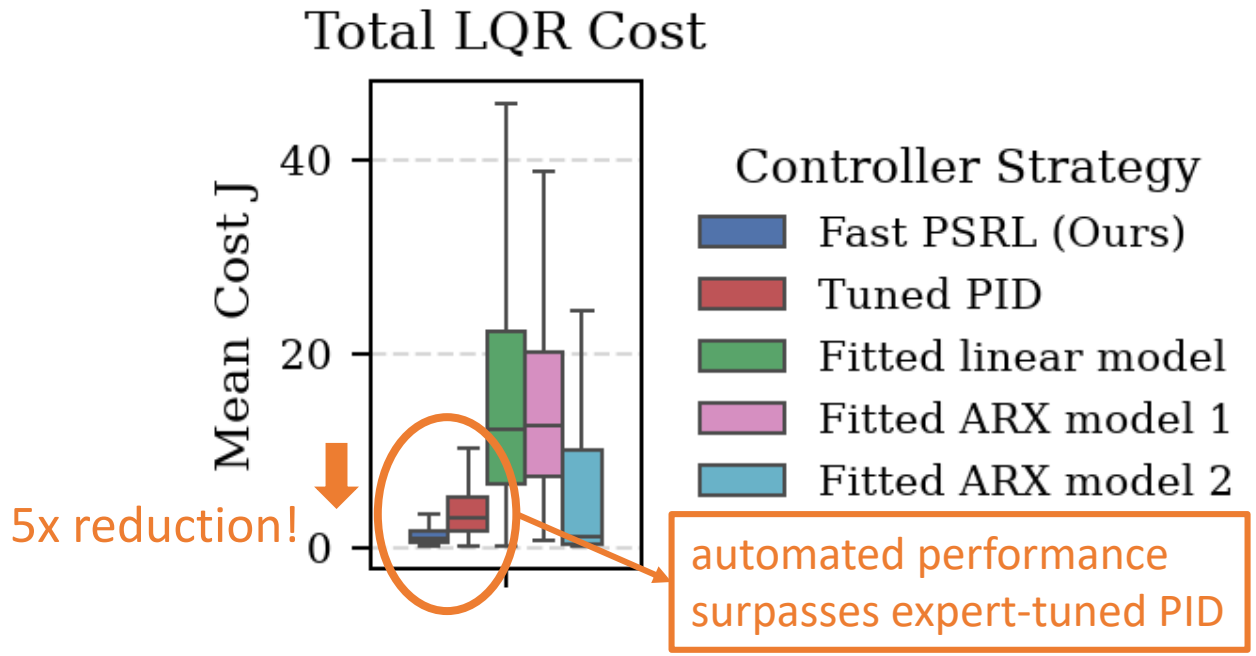


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G. Elmkaiel, M. Zaric, M. Muehlebach, "*Fast online reinforcement learning on real robots*", Conference on Robot Learning, 2026 (in review)



5x times lower cost than a expert-tuned PID++!

learned online with 120s of flight data!

Key difficulties

→ Continuous state/action systems

→ Nonlinear dynamics

→ Nonquadratic running cost

→ Uncountable parameter space

→ Non-episodic → identifiability/persistence of excitation is key!

$$\frac{1}{M}\sum_{k=1}^{M}\mathrm{E}\Big[|f^i(x_k,u_k)-f(x_k,u_k)|^2\ \Big|x_0\Big]\geq c_{\mathrm{e}}\sigma_{\mathrm{u}}^2$$

controllability of true dynamics

along trajectories

$$x_{k+1}=f(x_k,\mu^j(x_k)+n_{\mathrm{u}k})+n_k$$

$$n_{\mathrm{u}k}\sim\mathcal{N}(0,\sigma_{\mathrm{u}}^2 I)$$

M. Muehlebach, Z. He, M. I. Jordan, "*The Sample Complexity of Online Reinforcement Learning: A Multi-model Perspective*", International Conference on Learning Representations, 2026

Result

first finite-time non-episodic regret for continuous control!
matching lower bound → minimax optimal algorithm!

Theorem (parametric): Let the cost-to-go function corresponding to f and the stage cost be smooth, the feedback policies μ^θ be Lipschitz continuous, and let the persistence of excitation condition be satisfied. Then, for a constant learning rate η and $\sigma_{\mathrm{u}k}^2 \sim \sqrt{p/(d_{\mathrm{u}}k)}$, the policy regret of Alg. 2 is bounded by

$$\mathrm{E}[\sum_{k=1}^{T} l(x_k, u_k)] - T\gamma \leq \sqrt{(c_{\mathrm{r}1}\ln(T) + c_{\mathrm{r}2}p)d_{\mathrm{u}}T} + c_{\mathrm{r}3}\sigma^2 d_{\mathrm{x}},$$

for all $T \geq 2M$, where $c_{\mathrm{r}1}$, $c_{\mathrm{r}2}$, $c_{\mathrm{r}3}$ are explicit constants.

$$\gamma = \inf_{\mu} \lim_{T\to\infty} \frac{1}{T} \sum_{k=1}^{T} \mathrm{E}[l(x_k, u_k)]$$

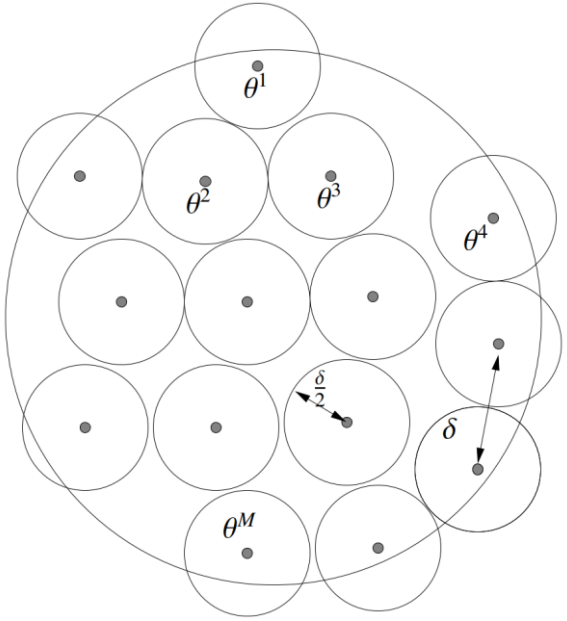
M. Muehlebach, Z. He, M. I. Jordan, "*The Sample Complexity of Online Reinforcement Learning: A Multi-model Perspective*", International Conference on Learning Representations, 2026

Discussion

- Packing/covering → broad class of dynamical systems
  - $\mathcal{O}(T^{(d_{\mathrm{x}}+d_{\mathrm{u}})/(d_{\mathrm{x}}+d_{\mathrm{u}}+2)}) \leq o(T)$ for nonlinear systems
  - $\mathcal{O}(\sqrt{T d_{\mathrm{x}} d_{\mathrm{u}}^2})$ for linear systems

- "Boundedness/asymptotic stability" for free:
  - Convergence in finite time (a.s.)
  - Bound on second moment: $\mathrm{E}[|x_k|^2] \leq c_{\mathrm{b}}$

- *Separation principle:*
  - Alg. 1 combines posterior sampling and certainty-equivalent control



M. J. Wainwright, "High-Dimensional Statistics", Cambridge University Press, 2019

M. Muehlebach, Z. He, M. I. Jordan, "*The Sample Complexity of Online Reinforcement Learning: A Multi-model Perspective*", International Conference on Learning Representations, 2026

Proof (finite number of models)

- Step I:

Proposition: Let the conditions of the main theorem be satisfied (persistence of excitation/smoothness). Then, the following holds:

$$\Pr(i_k \neq i^*) = \mathcal{O}\Big(\frac{1}{k^2}\Big),$$

where $f^{i^*} = f$.

- Intuition: $|x_{k+1} - f^i(x_k, u_k)|^2 = |f_k - f_k^i|^2 + 2n_k^{\mathrm{T}}(f_k - f_k^i) + n_k^{\mathrm{T}} n_k$

$$s_k^i - s_k^* = \sum_{j=1}^{k-1} |f_j - f_j^i|^2 + 2n_j^{\mathrm{T}}(f_j - f_j^i).$$

Martingale difference sequence

Proof (finite number of models)

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where $f^{i^*} = f$.

- Intuition:

$$\mathrm{E}\Big[e^{-\eta(s_k^i - s_k^*)}\Big] \approx e^{-\eta \mathrm{E}[s_k^i - s_k^*]}$$

Proof (finite number of models)

- Step II:
  - Lyapunov/value function $V : \mathbb{R}^{d_x} \to \mathbb{R}$ corresponding to $f = f^{i^*}, \mu = \mu^{i^*}$

→ $$\mathrm{E}[V(x_{k+1})] = \underbrace{\mathrm{E}[V(x_{k+1})|i_k \neq i^*]}_{\text{unfavorable}} \underbrace{\Pr(i_k \neq i^*)}_{\mathcal{O}(1/k^2)} + \underbrace{\mathrm{E}[V(x_{k+1})|i_k = i^*]}_{\text{favorable}} \Pr(i_k = i^*)$$

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$$\leq \mathrm{E}[V(x_k)] + \gamma + \mathcal{O}(\sigma_{\mathrm{u}k}^2) - \mathrm{E}[l(x_k, u_k)|i_k = i^*]$$

$$\leq c_2\mathrm{E}[V(x_k)] + \mathcal{O}(\sigma^2 + \sigma_{\mathrm{u}k}^2) - \mathrm{E}[l(x, u_k)|i_k \neq i^*]$$

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$$\leq (1 + \mathcal{O}(1/k^2))\mathrm{E}[V(x_k)] - \mathrm{E}[l(x_k, u_k)] + \gamma + \mathcal{O}(\sigma^2/k^2) + \mathcal{O}(\sigma_{\mathrm{u}k}^2)$$

→ Telescoping sum:

$$\sum_{k=1}^{T} \mathrm{E}[l(x_k, u_k)] - \gamma T \leq \mathcal{O}(\sigma^2) + \mathcal{O}(\sum_{k=1}^{T} \sigma_{\mathrm{u}k}^2) = \mathcal{O}(\ln(T) + \ln(m))$$

Meta-theorem:

Supervised learning has excess risk $\mathcal{O}(\sqrt{\log(m)/T})$, regret $\mathcal{O}(\sqrt{\log(m)T})$.*

Reinforcement learning **has the same rate** if:

1. Imitation learning/offline RL: distribution shift&simultaneity are controlled
2. Non-episodic online RL: models are identifiable/persistence of excitation
3. Decision-dependence: dynamics are contracting

*well known, see, e.g., [Robbins'51, Wolfowitz'54, Vapnik'95, Cesa-Bianchi'06, Bach'24, …]

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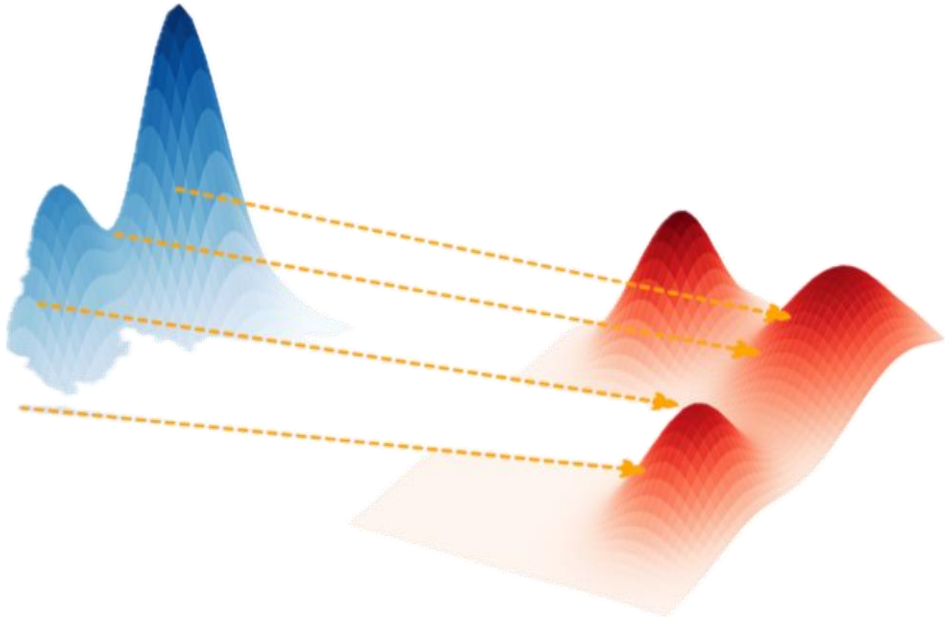
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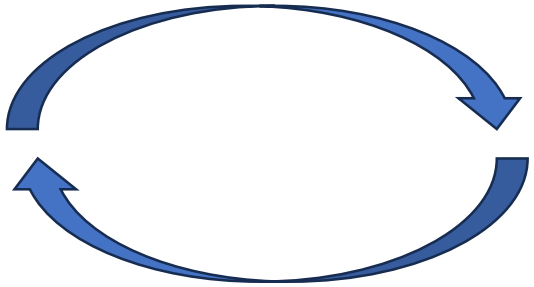
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Decision-dependent optimization

- The state / decision-variable is a probability measure when
  - using first-principle physics
  - probabilistic (Bayesian) view
  - distributional robustification
  - interaction with large population









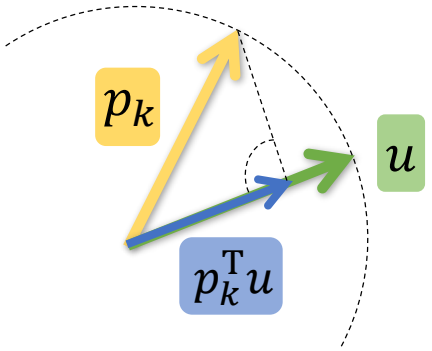
Z. He, S. Bolognani, F. Dörfler, M. Muehlebach, "*Decision-dependent stochastic optimization: The Role of Distribution Dynamics*", Journal of Machine Learning Research, 2026

Example: Polarized population model

Model [Gaitonde'21; Dean'22, Hazla'24]

$$p_{k+1} = \mathrm{norm.}(\ \lambda p_k + (1-\lambda)p_0 + \sigma(p_k^{\mathrm{T}} u_k)u_k\ )$$

individual position | initial position | biased assimilation: utility, ideology



$$|p_k| = |u_k| = 1$$

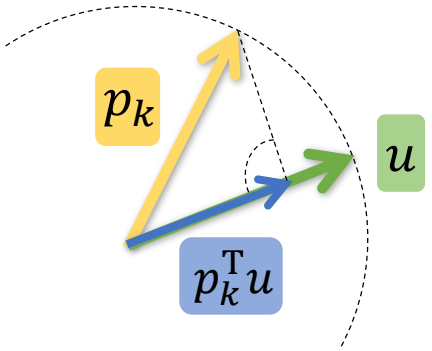
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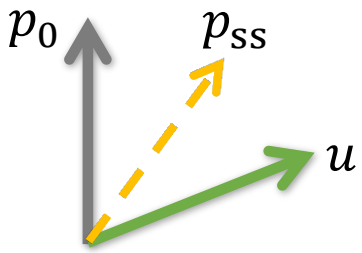
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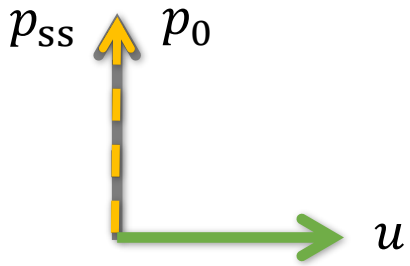


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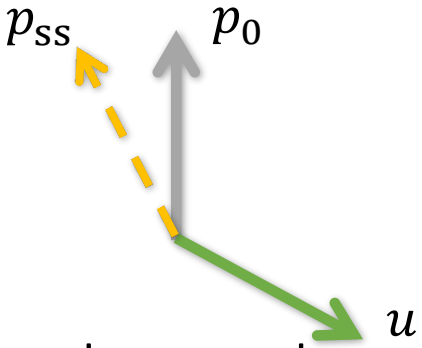
Steady-state $p_{\mathrm{ss}} = h(p_0, u)$ given a fixed ideological position u



acute angle



perpendicular angle



obtuse angle

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Modeling in probability space

- Model: $p_{k+1} = \mathrm{norm.}(\ \lambda p_k + (1-\lambda)p_0 + \sigma(p_k^{\mathrm{T}} u_k)u_k\ )$

$p_0 \sim \mu_0$

p_0 sampled from initial distribution μ_0

- Steady-state: $p_{\mathrm{ss}} = h(p_0, u)$ for fixed u

Z. He, S. Bolognani, F. Dörfler, M. Muehlebach, "*Decision-dependent stochastic optimization: The Role of Distribution Dynamics*", Journal of Machine Learning Research, 2026

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$p_0 \sim \mu_0$

- Steady-state: $p_{\mathrm{ss}} = h(p_0, u)$ for fixed u

p_0 sampled from initial distribution μ_0

- Distributional modeling: $\mu_{\mathrm{ss}}(u) = h(u, \cdot)_{\#}\mu_0$

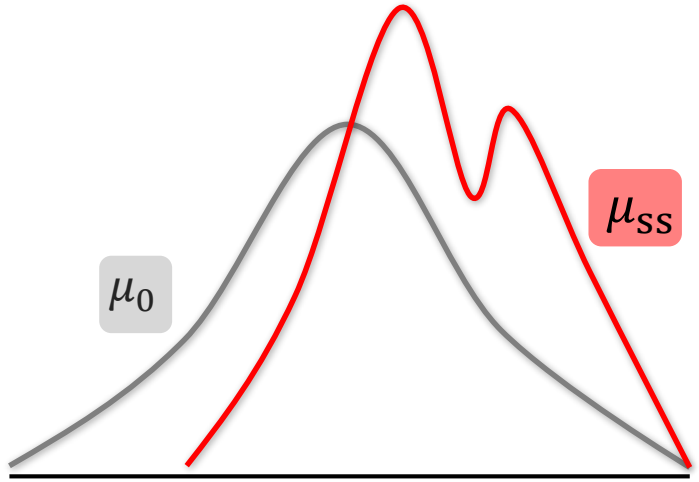
pushforward

- Fundamental assumptions:

$p_k = f(p_{k-1}, u_k, d), \qquad p_0 \sim \mu_0, \qquad d \sim \mu_d$

(contracting in p & Lipschitz in (d, u))

(finite absolute moments)



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Goal: Affinity maximization

- Distributional steady state: $\mu_{\mathrm{ss}}(u) = h(\cdot, u)_{\#}\mu_0$

- Goal:
$$\min_{|u|\leq 1} \quad \mathbb{E}_{p\sim\mu_{\mathrm{ss}}(u)}[\Phi(u,p)]$$
$$\text{s.t.} \quad \mu_{\mathrm{ss}}(u) = h(\cdot, u)_{\#}\mu_0$$



(population-wide affinity maximization)

- Challenges:
  - → nonlinear/unknown map h
  - → unknown μ_0
  - → system is not in steady-state
  - → decision-dependence $\mathrm{E}_{p\sim\mu_{\mathrm{ss}}(u)}$

decision-maker can at best sample from the currently observed μ_k

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Online stochastic algorithm

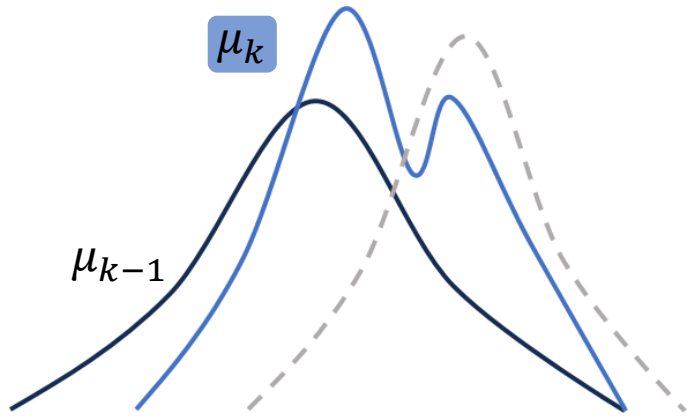
$$u_{k+1} = u_k - \frac{\eta}{n_{\mathrm{mb}}} \sum_{i=1}^{n_{\mathrm{mb}}} \underbrace{\big(\nabla_u \Phi(u_k, p_k^i) + \nabla_u h(p_0^i, u_k) \nabla_p \Phi(u_k, p_k^i)\big)}_{\text{composite stochastic gradient}}, \quad k \in \mathbb{N}$$

adaptation

anticipation & steering

… is a finite-sample approximation of

$$\mathbb{E}_{p_k \sim \mu_k}[\nabla_u \Phi(u_k, p_k) + \nabla_u h(p_0, u_k) \nabla_p \Phi(u_k, p_k)]$$



Z. He, S. Bolognani, F. Dörfler, M. Muehlebach, "*Decision-dependent stochastic optimization: The Role of Distribution Dynamics*", Journal of Machine Learning Research, 2026

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adaptation

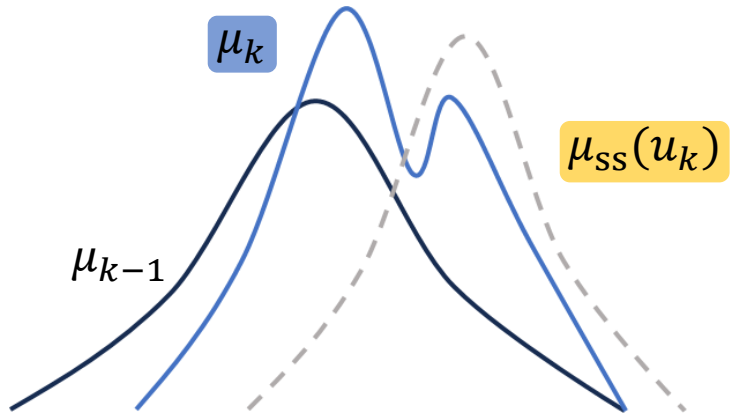
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$$\mathbb{E}_{p_k \sim \mu_k}[\nabla_u \Phi(u_k, p_k) + \nabla_u h(p_0, u_k) \nabla_p \Phi(u_k, p_k)]$$

… which approximates the gradient

$$\nabla \tilde{\Phi}(u_k) = \mathbb{E}_{p \sim \mu_{\mathrm{ss}(u)}}[\nabla_u \Phi(u_k, p) + \nabla_u h(p_0, u_k) \nabla_p \Phi(u_k, p)]$$

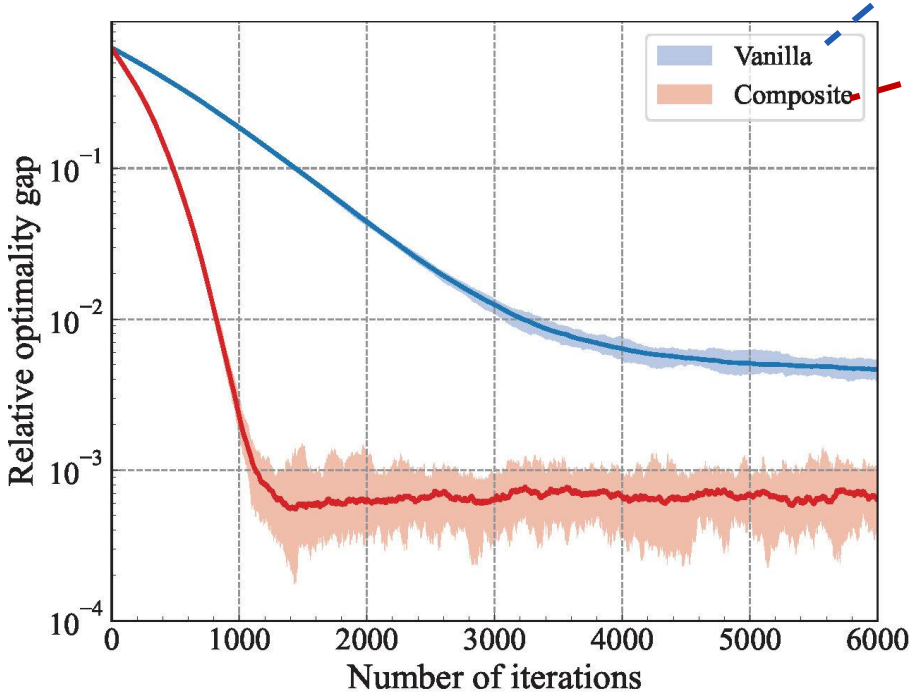


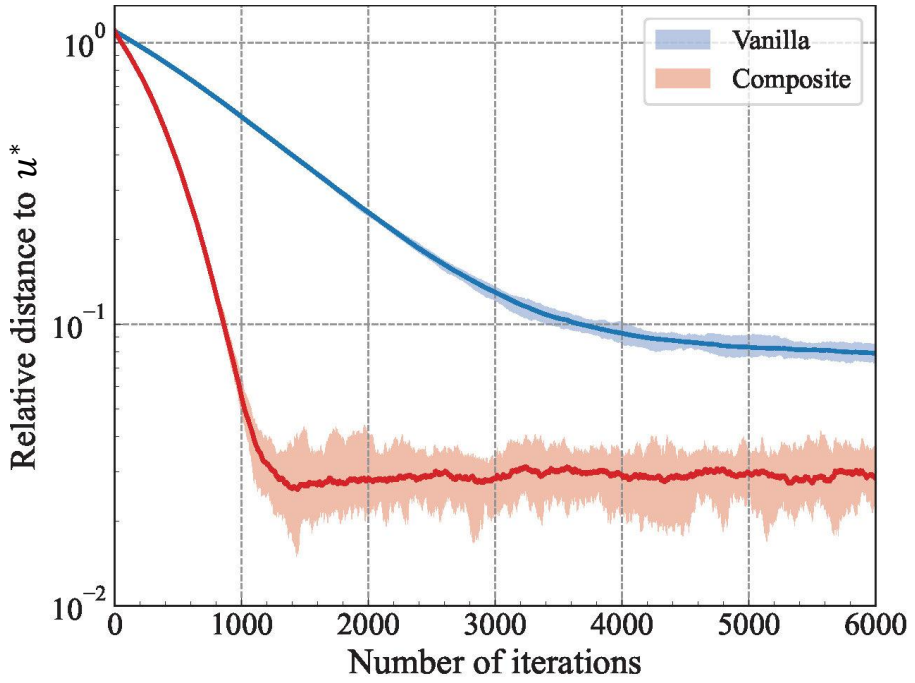
Z. He, S. Bolognani, F. Dörfler, M. Muehlebach, "*Decision-dependent stochastic optimization: The Role of Distribution Dynamics*", Journal of Machine Learning Research, 2026

Polarized population model

vanilla stochastic gradient descent without anticipation

our **composite** algorithm with anticipation & sensitivity estimation

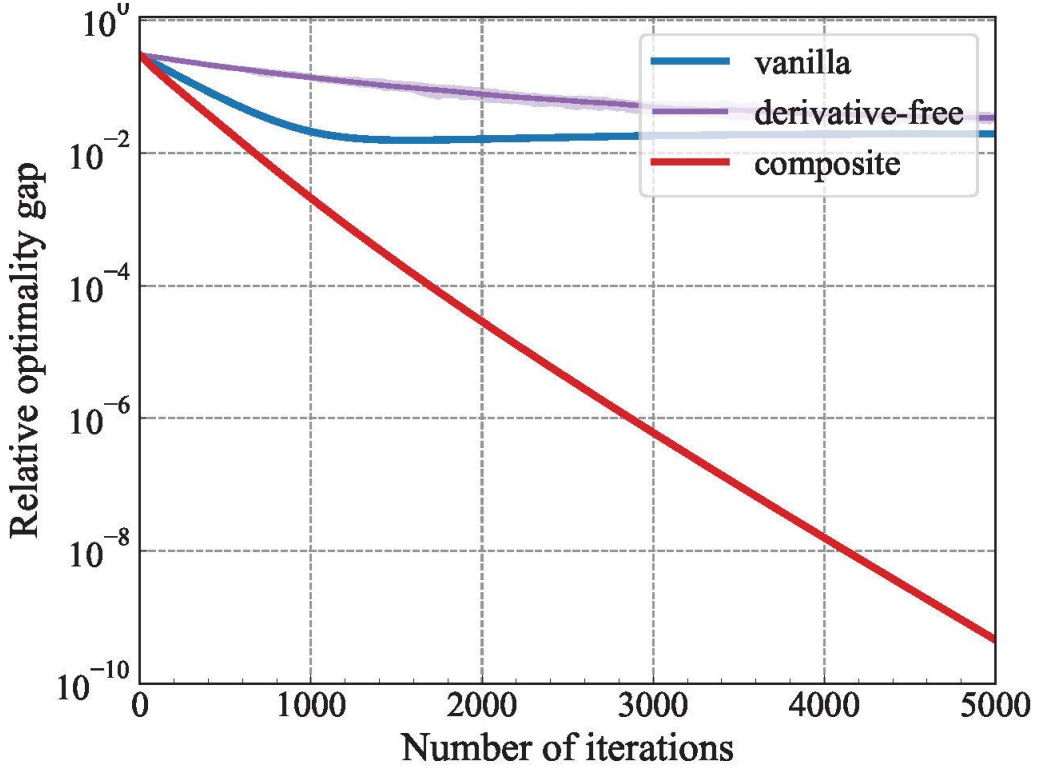


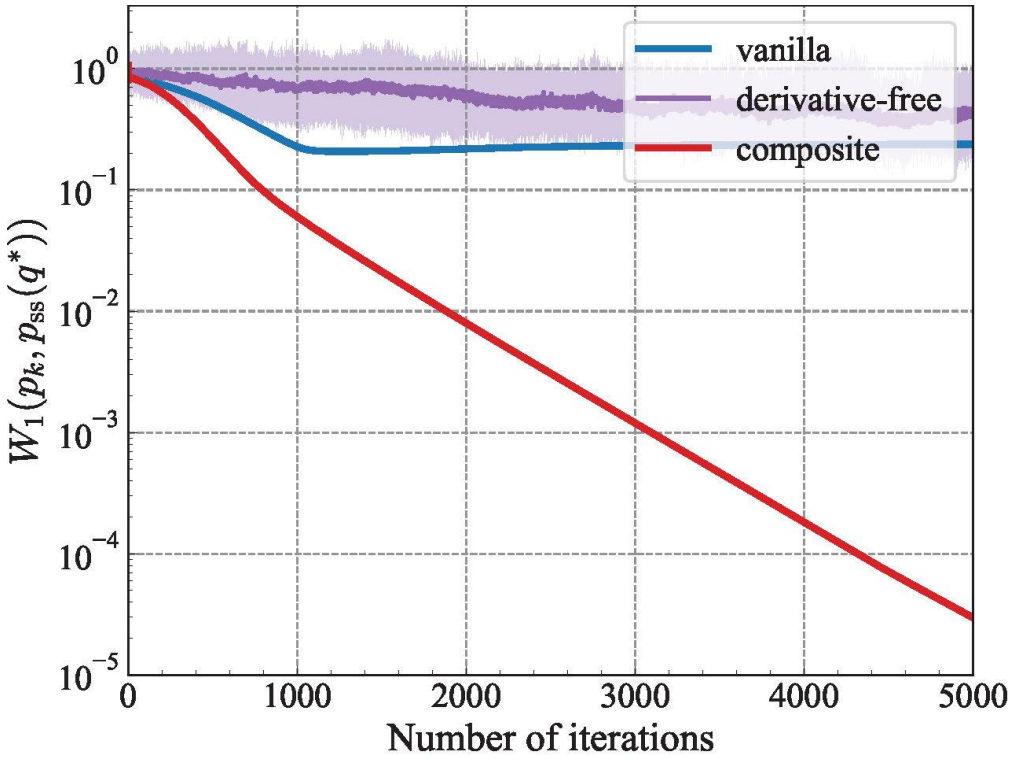


Z. He, S. Bolognani, F. Dörfler, M. Muehlebach, "*Decision-dependent stochastic optimization: The Role of Distribution Dynamics*", Journal of Machine Learning Research, 2026

Recommender system

- Recommender systems exhibit feedback and dynamics! [Chaney'18, Mansoury'20,Perdomo'20,Kraut'22]



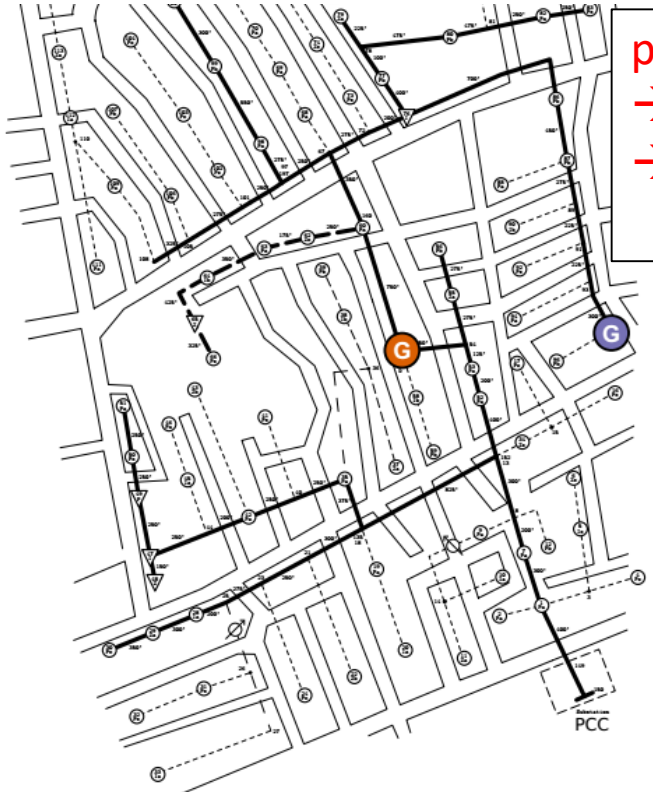


Z. He, S. Bolognani, F. Dörfler, M. Muehlebach, "*Decision-dependent stochastic optimization: The Role of Distribution Dynamics*", Journal of Machine Learning Research, 2026

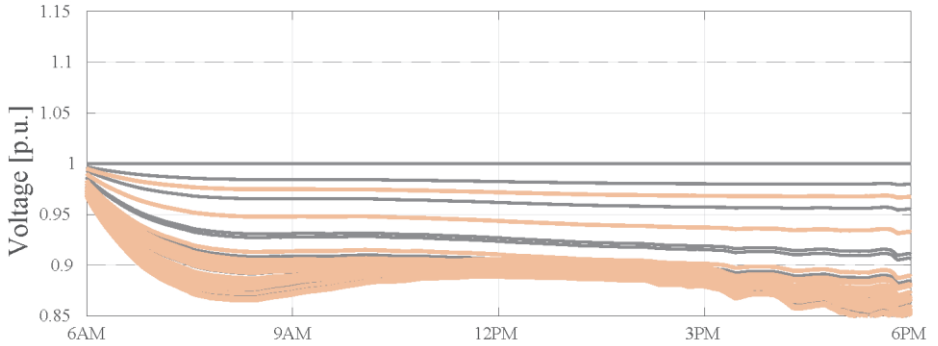
Example: power-grid

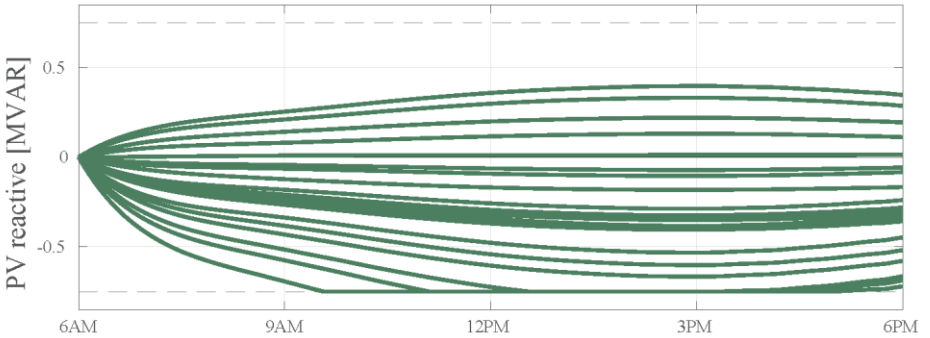
UNICORN – 56-bus test scenario

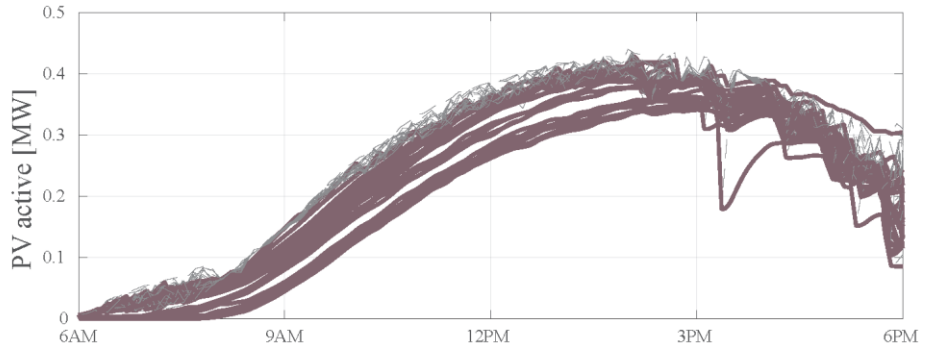
L. Ortmann, A. Hauswirth, I. Caduff, F. Dörfler, S. Bolognani, "*Experimental validation of feedback optimization in power distribution grids",* Electric Power Systems Research, 2020



power distribution network:
→ 15 PV generators are added
→ control reactive/active power of PV generators







W. Chan*, Z. He*, K. Moffat, S. Bolognani, M. Muehlebach, F. Dörfler, "Robust Feedback Optimization with Model Uncertainty: A Regularization Approach", IEEE Conference on Decision and Control, 2025

Optimal decision-making

Despite dynamics & feedback structure we achieve the same (optimal) rate as SGD!

Theorem: The decision-making algorithm with step $\eta = \mathcal{O}(1/\sqrt{T})$ guarantees:

$$\frac{1}{T}\sum_{k=0}^{T-1} \mathbb{E}\left[|\nabla\tilde{\Phi}(u_k)|^2\right] \leq \frac{8(\tilde{\Phi}(u_0) - \mathrm{E}[\tilde{\Phi}(u_T)])}{\eta T} + \frac{4LM\eta}{n_{\mathrm{mb}}} + \mathcal{O}\left(\frac{1}{T}\right)$$

$$= \mathcal{O}\left(\tfrac{1}{\sqrt{T}}\right)$$

where $\tilde{\Phi}(u) = \mathbb{E}_{p\sim\mu_{\mathrm{ss}}(u)}[\Phi(u,p)]$.

Moreover, it holds with probability $1-\delta, \delta \in (0,1)$,

$$\frac{1}{T}\sum_{k=0}^{T-1} |\nabla\tilde{\Phi}(u_k)|^2 = \mathcal{O}\left(\frac{1}{\sqrt{T}}(1+\log(1/\delta)\right)$$

$$\frac{1}{T}\sum_{k=0}^{T-1} W_1(\mu_k, \mu_{\mathrm{ss}}(u_k)) = \mathcal{O}\left(\frac{1}{T^{1/4}}\sqrt{1+\log(1/\delta)}\right).$$

Z. He, S. Bolognani, F. Dörfler, M. Muehlebach, "*Decision-dependent stochastic optimization: The Role of Distribution Dynamics*", Journal of Machine Learning Research, 2026

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Is supervised ≈ reinforcement learning?



Theme	Contribution	Observational learning if
imitation learning	quantization-free transformer bilevel optimization	no guarantee (distribution shift/simul. bias!)
non-episodic online RL	freq. regret bounds continuous state/action	identifiability/pers. of ex. realizability
decision-dep. learning	online opt. in feedback	contraction

Z. Sheebaelhamd, M. Tschannen, M. Muehlebach, C. Vernade, "*Quantization-Free Autoregressive Action Transformers*", Advances in Neural Information Processing Systems, 2025

Z. He, S. Bolognani, F. Dörfler, M. Muehlebach, "*Decision-dependent Stochastic Optimization*", arXiv:2503.07324, 2025

M. Muehlebach, Z. He, Michael I. Jordan, "*On the Sample Complexity of Online Reinforcement Learning: A Multi-model Perspective*", arXiv:2501.15910, 2025